

## Assignment 6

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### Triage Pseudocode

1. After the patient enters the ED, the attending nurse/physician comes to them with a hospital owned tablet containing the triage model.
2. The medical professional immediately observes the patient. If immediate life-threatening conditions such as a heart attack or a lack of breathing are observed, the patient is immediately rushed for treatment without going through the rest of the digital triage process.
3. If no immediate life-threatening conditions are observed the patient is assessed for: Age, Gender, GCS, SBP, DBP, Pulse, Temperature, Respiratory Rate, SpO2 and purpose for visit. Shock index and MAP are calculated from these values; then, an ID number is assigned to the patient.
4. These attributes are fed to a pre-trained ML/DL algorithm running locally on the device. The information is then stored on the hospital's records in a secure database.

**Table showing some metrics/criterion used to classify triage levels**

| <b>Metrics/Criterion</b>                                                                                                                                                                                                                      | <b>Triage level</b> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Patient in immediate risk of death (skip rest of tests and treat immediately).                                                                                                                                                                | Level 1             |
| Patient has:<br><b>GCS</b> 9–12 (diminished mental status but can protect airway)<br><br><b>SBP</b> : 90–100 mmHg<br><br><b>DBP</b> : 60–70 mmHg<br><br><b>Pulse</b> > 100 bpm<br><br><b>Temperature</b> : 38.5–40°C (dangerously high fever) | Level 2             |

|                                                                                                                                                   |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <b>Respiratory Rate</b> > 20<br><br><b>SpO<sub>2</sub>:</b> 90–94% (moderate hypoxia)<br><br><b>MAP:</b> 65–75 mmHg (mild perfusion difficulties) |         |
| Patient suffering from immediate acute illness/injury but otherwise stable.                                                                       | Level 3 |
| Patient needs a minor procedure. ie stitches.                                                                                                     | Level 4 |
| Patient is in for routine reason                                                                                                                  | Level 5 |

5. A triage score is assigned to the patient by the ML/DL algorithm. The patient then receives an identification band.
6. The patient is escorted to the waiting area and is seen according to their triage level.